STAGE 1 / SESSION 03

Dataset screening & cleaning

CricketLens | Men's IPL EDA + two future probabilistic prediction models

Decision	Current status
Stage 1	Completed for EDA, with documented source-data exceptions
Source	Cricsheet men's IPL JSON archive provided for the project
Scope	Seasons 2008-2026; 2026 reserved from ML training and tuning

1 / Verified scope and output

Data quality / output	Measured result
Match JSON files	1,243 men's IPL matches
Ordinary innings / deliveries	2,480 innings; 295,557 ball-by-ball records
Batter innings	18,768 player batting-innings records
Next-over candidate samples	44,706 snapshots; not final ML training eligibility
Source identity and integrity	0 duplicate match IDs; 0 delivery run-total mismatches; 0 negative run fields
Totals reconciliation	0 innings run/wicket/legal-ball reconciliation failures
Tests	9 automated tests passed on synthetic fixtures

2 / Key corrections in this session

Added a source-level JSON audit and conservative venue standardization. Both original venue and venue_canonical remain available. Candidate next-over training checkpoints after a corrupted over with more than six legal deliveries are excluded, rather than using unreliable cumulative score features.

Venue identity	Measured result
Source venue labels	60 distinct raw strings
Reviewed canonical labels	38 distinct standardized venue names
Alias changes	22 specific raw names standardized, affecting 461 matches
Caution	No unverified merging of redeveloped or potentially different cricket grounds

STAGE 1 / AUDIT DECISIONS

What we discovered and how we handled it

Finding	Treatment / interpretation
51 missing city fields	33 Dubai and 18 Sharjah records; retained blank instead of guessing/imputing. Missingness mechanism not proven.
25 missing winners	16 tied matches and 9 no-results. These are structurally undefined winners, not random corruptions.
23 D/L matches	Flagged for explicit model-cohort eligibility decision in Stage 3; 608 candidate next-over checkpoints come from them.
4 overs containing 7 legal balls	Source anomalies retained and documented. Later training snapshots from affected innings excluded.
4 overs with 0 legal balls	Legitimate last-over winning wides in second-innings chases; kept as source facts.
6 one-innings matches	Retained as source-coverage exceptions; no invented missing innings.
34 Super Over innings	Excluded from ordinary-innings analysis; not merged into scheduled 20-over innings.
Player-name ambiguity	"Harmeet Singh" appears with two different player IDs; player features must use registry IDs.
Censored next-over windows	Incomplete next overs without all-out are not treated as observed negative labels.

3 / What is NOT finished

Data preparation is complete enough for systematic EDA, but the ML training cohort is not frozen. Stage 3 must define handling of rain-affected matches, incomplete next-over windows, past-only pre-match player features, and possible sample-selection bias. Earlier 2025 baseline metrics belong to a superseded provisional extraction and must be re-evaluated.

4 / GitHub and local data

GitHub ZIP: updated Python parser, review code, nine tests, documented exceptions, development log and this report. Processed-data ZIP: revised local CSV tables only. The raw IPL ZIP, derived CSVs and trained model files are ignored by Git.

5 / Next phase: Stage 2 - Exploratory Data Analysis

Next we will produce reproducible descriptive statistics (mean, median, spread, outliers), missingness visualizations, score and wicket distributions, team/player and venue comparisons, season trends, Pearson/Spearman relationships, and written interpretations grounded in the data. No additional prediction models are being added.

Documentation: docs/STAGE_01_STATUS.md | docs/STAGE_01_SESSION_03_FINDINGS.md | reports/stage_01/*.csv and *.json